

CSA0711

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**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Code of Conduct:

*I Purnima / 192312642 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%				
Method Selection	20%				
Solution Process	25%				
Accuracy of Calculation	20%				
Interpretation of Results	15%				

Total Marks 26 /40

Signature of the Faculty

1. A university network is assigned the IP address block 192.168.20.0/24

Given:-

Solution:-

main network : 192.168.20.0/24

Lab A : /26

Lab B : /27

Lab C : /28

Lab A (/26)

Subnet mask : 255.255.255.192

Block Size : 64 addresses

Network addresses : 192.168.20.0/26

Item	value
Network address	192.168.20.0
First Host	192.168.20.1
Last Host	192.168.20.62
Broadcast address	192.168.20.63
Total address	64
usable	62

Step 1: lab 8 (7/7)

subnet mask : 255.255.255.224

Block size : 32 addresses

network address : 192.168.20.64/27

Item	Value
network address	192.168.20.64
first host	192.168.20.65
last host	192.168.20.94
broadcast address	192.168.20.95
total addresses	32
usable hosts	30

Step 3: lab 6 (7/8)

subnet mask : 255.255.255.240

block size : 16 addresses

Item	Value
network address	192.168.20.96
first host	192.168.20.97
last host	192.168.20.110
broadcast address	192.168.20.111
total addresses	16
usable hosts	14

2. A network administration :-

System 1 :- IP address 192.168.1.130 Subnet mask : 255.255.255.192 (7/26)

System 2 :-

IP address : 192.168.1.190

Subnet mask : 255.255.255.192 (7/26)

Step 1 :-

Subnet mask = 256 - 255 = 1

Block size = 256 - 192 = 64

Network address	Host Range	Broadcast address
192.168.1.0	192.168.1.1 - 192.168.1.62	192.168.1.63
192.168.1.64	192.168.1.65 - 192.168.1.126	192.168.1.127
192.168.1.128	192.168.1.129 - 192.168.1.190	192.168.1.191
192.168.1.192	192.168.1.193 - 192.168.1.254	192.168.1.255

System 1 analysis :-

IP address : 192.168.1.130

Since 130 lies between 128 and 191, it belongs to the third subnet

Item	Value
network address	192.168.1.128
first host	192.168.1.129
last host	192.168.1.190
broadcast address	192.168.1.191
IP valid ?	Yes

3. An organization has been allocated :-

Network :-

Mask :-

network block : 172.16.0.0/16

Department and subnet masks :-

Department and subnet masks.

HR / 25 Sales / 26 IT / 27 admin / 28

1. HR department (25)

Subnet mask :- 255.255.255.128

Block size : 128 addresses.

Items:-	Values
network address	172.16.0.0
First host	172.16.0.1
Last host	172.16.0.126
Broadcast address	172.16.0.127
Total address	128
usable Hosts	126

2. Sales department (/26):-

Subnet mask : 255.255.255.192

Block Size : 64 address.

3. IT department (/27):-

Subnet mask : 255.255.255.192

Block Size : 32 addresses

4. Admin department (/28):-

Subnet mask : 255.255.255.240

Block size : 16 addresses

4. A company uses the IP address:-

Given:-

network addresses : 172.16.10.0/24

calculate total no. of subnets

original network = /24

new subnet mask = /27

no. of borrowed bits :- $27 - 24 = 3$

Total no. of subnets :-

$2^3 = 8$ subnets

calculate usable hosts per subnet :-

Remaining host bits :- $32 - 27 = 5$

Tot. addresses per subnet :- $2^5 = 32$

usable host address :- $32 - 2 = 30$

determine the subnet of 172.16.10.78

Subnet 172.16.10.64/27

network addresses 172.16.10.64

valid Host Range : 172.16.10.65 - 172.16.10.94

Broadcast addresses : 172.16.10.95

5. IPV6 addressing :-

given :-

2001:0db8:0000:0000:0000:ff00:0042:8329

compressed form :-

2001:db8:ff00:42:8329

Expand from :-

2001:0db8:0000:0000:0000:0000:ff00:0042:8329

Network prefix (/64)

2001 : adb8 : 0000 : 0000 : /64

number of Host addresses

IPv6 host bits

$$128 - 64 = 64$$

Total host addresses

$$2^{64} = 181440744073709551616$$